

DATE: _____

DAY: _____

Group - 2

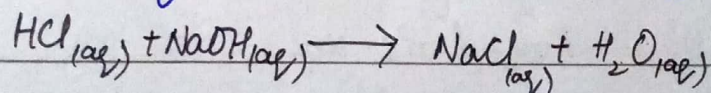
Past Paper - 2022

—: (i): —

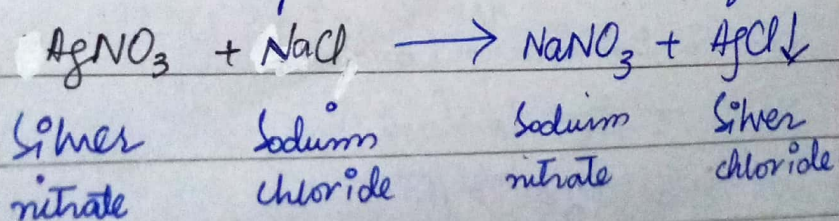
Ionic Reactions

Ionic reactions are generally the reaction of ions that are anions & cations to form a compound in a medium in which they are dissolved. When ions of water-soluble salts interact each other in aqueous medium results in the formation of water insoluble salts.

For Example:- When we mix HCl with aqueous NaOH it results in formation of salt NaCl & water H_2O .



Another example is net reaction for.



DATE: _____

DAY: _____

DATE: _____

—: () :—

Barometric Formula

It is also known as exponential atmosphere or Isothermal atmosphere.

It is a formula we use to model how the pressure or density of the air changes with altitude.

The pressure drops approximately by 11.3 pascals per meter in the first 1000 meters above the level of the sea.

A Barometric formula gives the pressure in planetary atmosphere usually the Earth's as a function of height.

Derivation:-

$$P = \frac{\text{mass}}{\text{Volume}} = \frac{n N_A m}{n R T / P}$$

$$\therefore \frac{R}{N_A} = k$$

n = no. of moles

N_A = Avogadro's No.

m = mass of One Molecule

k = Boltzmann Constant

R = Gas Constant.

$$P = P_0 e^{-mgh/RT}$$

DATE: _____

DAY: _____

—: (iii):—

Cage Effect

Cage effects describe how the properties of a molecule are affected by its surroundings.

It is also known as Geminate Recombination.

In solvent, a molecule is often more accurately described existing in a cage of solvent molecules, the so called solvent cage.

In a gas at low pressure a normally highly reactive & short lived molecule can be contained specific host-guest interactions occur in inclusion compounds.

In solution, solvent is a major factor in kinetics. Limited proximity of reactants. Molecules must diffuse into reaction zone.

—: (iv):—

Effective collisions

An effective collision is the one in which molecules collide with sufficient energy

↓ proper Orientation. This will happen when

- the reactant species collide
- colliding molecules have total kinetic energy equal to or greater than activation energy of reaction.
- colliding molecules must have proper orientation relative to each other.

Effective collisions result in product formation. Ineffective collisions do not result in product formation.

In Arrhenius equation, fraction of collision is given by $e^{-E_a/RT}$ as P represents the fraction of molecules with energies equal to or greater than E_a .

—————: (v): —————

Pre-Exponential Factor

Arrhenius proposed an empirical equation for calculating the Activation energy of a reaction having rate constant k at Temperature T .

$$k = Ae^{-E_a/RT}$$

DATE: _____

DAY: _____

E_a = Arrhenius Activation Energy.

A = Pre-exponential factor

R = Gas constant.

E_a & A known as Arrhenius parameters

Taking log On B.S.

$$\ln k = \ln(Ae^{-E_a/RT})$$

$$\ln k = \ln A - \frac{E_a}{RT}$$

$$\ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A.$$

$$y = mx + c$$

We can calculate pre-exponential factor by plotting a graph $\ln k$ against corresponding reciprocal temperature

When we plot $\ln k$ against $\frac{1}{T}$, a straight line is obtained with slope equal to $\frac{E_a}{R}$ and an intercept which is equal to $\ln A$.

If we take the exponential of $\ln A$ which is equal to A .

—————: (vi) :—————

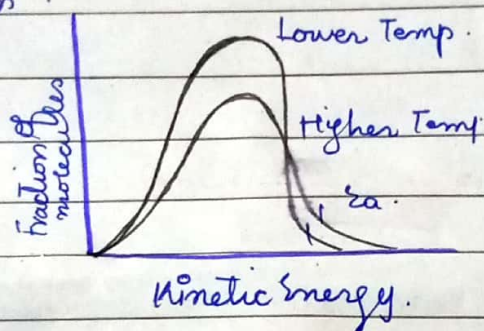
Effect of Temperature

An increase in Temperature typically increases the rate of reaction. An

DATE: _____

DAY: _____

increase in temperature will raise the average kinetic energy of reacting molecules. Therefore, a greater proportion of molecules will have minimum energy necessary for an effective collision.



—: (vii):—

Clausius Inequality

Clausius inequality for a closed system states that the process is irreversible if $TdS > dQ$ and process is Reversible if $TdS = dQ$.

It applies to any real engine cycle and implies a negative change in entropy on the cycle.

(If any amount of energy added by heating) can be measured during the process & the temperature can be

DATE: _____

measure
Clausius
can be
the pro
sible.

in U

A p
inequa
in

It

that

nam

sys

con

Th

P

S

DATE: _____

DAY: _____

measured from the process. then Clausius inequality will be used or can be used to determine whether the process is reversible or irreversible. by carrying out the integration in Clausius inequality

A practical example of Clausius inequality is the use of compressor in a refrigerator system.

—————: (VIII): —————

Partition function

It is an important parameter that is link between the thermodynamic coordinates of a macroscopic system and coordinates of the constituent microscopic system.

The system Partition function for a particle in Mobile phase is of general form.

$$Q = q, N N! = (q_{\text{trans}} q_{\text{rot}} q_{\text{vib}} q_{\text{elec}}) N N!$$

Factorization of Partition Function.

Distribution of energy among a large no. of molecules (N). within a con-

iner of Volume. that is maintained (by) ⁱⁿ equilibrium by such energy sharing. at specified temperature T is the most important equation in statistical mechanics

—————: (ix): —————

Nernst Heat Theorem

Any system undergoing a process between equilibrium states as a result of external influences such as Pressure, experience as change in Entropy.

This change in entropy tends to zero as the temperature characterizing the process tends to absolute zero.

One of the application of Third law of Thermodynamics or Nernst Heat Theorem is that it can be used to determine absolute entropies of different substances at any T .

The statement of Nernst was simplified by Planck.

DATE: _____

DAY: _____

— :: (X) :: —

Root Mean Square Velocity

Crms:- Root mean square Velocity

Mean Square Velocity is defined as.

$$\bar{c^2} = \frac{c_1^2 + c_2^2 + c_3^2 + \dots + c_n^2}{n}$$

$$\bar{c^2} = \frac{1}{n} \sum c_i^2$$

$$\bar{c^2} = \frac{1}{n} \int_0^\infty c^2 dn_c \quad \text{--- (1)}$$

$$\bar{c^2} = \frac{\int_0^\infty c^2 \frac{dn_c}{n}}{\quad} \quad \text{--- (1')}$$

Maxwell Law:-

$$\frac{dn_c}{n} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-mc^2/2kT} c^2 dc$$

Put this in (1')

$$\bar{c^2} = \int_0^\infty c^2 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-mc^2/2kT} c^2 dc$$

$$= 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^\infty e^{-mc^2/2kT} c^4 dc$$

Let Suppose:- $\frac{m}{2kT} = \frac{1}{\alpha^2}$

$$\bar{c^2} = 4\pi \left(\frac{1}{\alpha^2 \pi} \right)^{3/2} \int_0^\infty e^{-c^2/\alpha^2} c^4 dc$$

We know:-

DATE: _____

DAY: _____

$$\int_0^{\infty} e^{-c^2/\alpha^2} c^4 dc = \frac{3}{8} \alpha^5 \sqrt{\pi}$$

$$\bar{c}^2 = 4\pi \left(\frac{1}{\alpha^3 \pi} \right)^{3/2} \frac{3}{8} \alpha^5 \sqrt{\pi}$$

$$\bar{c}^2 = \frac{4\pi}{\alpha^3 \pi^{3/2}} \frac{3}{8} \alpha^5 \pi^{1/2}$$

$$\bar{c}^2 = \frac{3}{2} \alpha^2$$

Taking Underroot.

$$\sqrt{\bar{c}^2} = \sqrt{\frac{3}{2} \alpha^2}$$

$$c_{rms} = \sqrt{\frac{3}{2}} \alpha$$

$$\because \frac{1}{\alpha^2} = \frac{m}{2kT}$$

$$\alpha = \sqrt{\frac{2kT}{m}}$$

$$c_{rms} = \sqrt{\frac{3}{2}} \sqrt{\frac{2kT}{m}}$$

$$= \sqrt{\frac{3kT}{m}}$$

$$\because N_A m = M$$

$$c_{rms} = \sqrt{\frac{3RT}{m N_A}}$$

$$\because k = \frac{R}{N_A}$$

$$c_{rms} = \sqrt{\frac{3RT}{M}}$$

$$c_{rms} = 1.732 \sqrt{\frac{RT}{M}}$$

DATE: _____

DAY: _____

_____ ? (XI) ? _____

Classical Thermodynamics

It is a description of Thermodynamic system states at near equilibrium using macroscopic measurable properties

It concerns the relationship b/w bulk properties of Matter. Nothing is examined at atomic or molecular level.

It is used to stimulate energy, work & heat exchanges using Thermodynamic Laws.

Statistical Thermodynamics

It is theory that predicts the behaviour of Microscopic quantities of compounds based on molecular properties

It seeks to explain those bulk properties in terms of constituent atoms. It treats the aggregation of atoms.

It has been used to predict reaction rates in general.

(xii) ————— Nernst Approximation Formula

Nernst equation relates the instantaneous potential E to standard potential E° & the reaction quotient Q .

$$E = E^\circ - (0.0592 \text{ V}/n) \log Q \text{ at } 298\text{K}.$$

When all reactants & products are in their standard states.

$$Q = 1 \quad \text{and} \quad E = E^\circ.$$

(xiii) Pulse Method.

(xiv) Fast Reactions

(xv) Postulates of Bimolecular Theory.

Long Question

Eyring Equation & its Parameters

Effect of T & Altitude or Vertical distribution constant.

Expression of E in terms of Partition Function

Third Law of Thermodynamics.

Pg. 239.

Transition State Theory Postulates.

Derive Two Maxwell Relation.